



Oxidation effect on pool boiling critical heat flux enhancement of Cr-coated surface for accident-tolerant fuel cladding application

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ABSTRACT

The oxidation of an accident-tolerant fuel (ATF) cladding alters the surface roughness, which, in turn, affects the capillary wicking characteristics and critical heat flux (CHF) during the boiling process. To evaluate the effect of oxidation on the boiling performance of ATF coating, this study fabricated Cr-coated surfaces by varying the substrate temperature, resulting in various sizes of deposited particulate nanostructures by using a direct-current magnetron sputtering technique. To consider the oxidation effect, sputtered test specimens were oxidized at 400 °C dry air for 20 days, forming a Cr₂O₃ oxide layer with a thickness of approximately 0.1 μm. Pool boiling CHF experiments were conducted in deionized water under atmospheric pressure. The resultant CHF on the fresh and oxidized Cr-coated surfaces was enhanced by up to 36% and 48%, respectively. A higher CHF enhancement was achieved with a lower substrate temperature during sputtering, which resulted in a higher nanoscale surface roughness. The spreading velocities on each specimen were measured through the characterization experiments of liquid droplet spreading, which can be interpreted as the capillary wicking momentum. The pool boiling CHF was predicted with spreading velocity and roughness parameters. The predicted CHF showed a good agreement with the measured CHF. This study suggests that the oxidation of Cr-coated ATF specimens has a positive effect on the enhanced CHF, which can be confirmed through the altered surface roughness and capillary wicking momentum.

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1. Introduction

Since the Fukushima Daiichi nuclear power plant (NPP) accident in March 2011, tremendous efforts have been invested for enhancing the safety of NPP. Through the numerous activities conducted for enhancing nuclear safety, system scale applications such as the passive safety injection system, passive containment cooling system, and core catcher system have been developed and their deployment is underway. Besides the safety facilities in the system scale, an innovative concept of accident tolerant fuel (ATF) cladding was recently suggested as a practical way to inherently improve the nuclear reactor safety. ATF cladding materials, in principle, passively offer a prolonged coping time under the accident conditions. In particular, the ATF cladding significantly reduces hydrogen generation during a high-temperature steam-cladding reaction compared to the current zirconium-based claddings [1]. As a developing strategy for the ATF cladding, a deposition of the

ATF cladding materials onto conventional zirconium-based cladding is preferred owing to its shorter development time and minimal differences in the thermo-mechanical and neutronic properties compared to conventional cladding [2,3].

For a practical application within the nuclear reactor, however, the boiling performance, as well as the aforementioned characteristics of the developing ATF candidates, needs to be investigated thoroughly. In particular, it has been revealed that a change in surface micro/nanoscale features possibly generated from the depositing techniques can affect the critical heat flux (CHF), which plays a critical role in determining thermal safety margin in nuclear reactor core [4–7]. A cause of the enhancement in CHF has been widely understood by capillary wicking mechanism. In the mechanism, the micro/nanoscale structures enhance the capillary action that effectively delays a growth of dry spots/patches, which leads to the CHF enhancement. These results have been demonstrated by in-situ direct observations, in which a growth of dry spots was significantly delayed by a capillary liquid inflow [8,9]. As a parameter representing the capillary wicking, surface roughness is widely used because the increase in the surface roughness enlarges liquid/solid contact length, and thereby amplifying the capillary

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Nomenclature

R_a	arithmetical mean height (nm)	A_t	total heat transfer area (m ²)
x	thickness of oxide layer (μm)	A_c	total cross-sectional area of wicked channel (m ²)
k	adjustment constant (–)	$\dot{V}_w$	wicked liquid volume rate (mm ³ /s)
Δm	mass gain after oxidation process (mg)	D	contact diameter of liquid droplet (mm)
A_{ox}	activating area for oxidation (m ²)	D_s	contact diameter at transient point (mm)
ρ_{ox}	density of Cr ₂ O ₃ (kg/m ³)	t	spreading time (ms)
M_{ox}	molar mass for Cr ₂ O ₃ (g/mol)	t_s	spreading time at transient point (ms)
M_{O_2}	molar mass for O ₂ (g/mol)	D_b	bubble base diameter (mm)
R_{sm}	mean width of the profile element (μm)	K	adjustment factor in this study (–)
r_n	nanoscale surface area (–)		
q''	critical heat flux (kW/m ²)		
Power	inserted power to specimen (W)		
A_{heated}	heated area in specimen (m ²)		
V	voltage (V)		
I	current (A)		
W	width of heat transfer area (m)		
L_{heated}	heated length (m)		
$U_{q''}$	error of measurement in heat flux (kW/m ²)		
U_V	error of measurement in voltage (V)		
U_I	error of measurement in current (A)		
U_W	error of measurement in width of heat transfer area (m)		
$U_{L_{heated}}$	error of measurement in heated length (m)		
S	adjustment factor in literature (–) [9]		
h_{fg}	latent heat (kJ/kg)		
g	gravitational acceleration (m/s ²)		
C	constant (–) [9]		
S_m	mean spacing (μm)		
$\dot{m}_w$	mass flow rate of wicked liquid inflow (kg/s)		
u_w	wicking velocity (mm/s)		

Greek letters

ρ	density (kg/m ³)
σ	surface tension (N/m)
θ	contact angle (°)

Subscripts

ox	oxide
CHF	critical heat flux
v	vapor
l	liquid
t	total
w	wicking
c	channel
s	transient point of spreading behavior
micro	microscale (without nanoscale consideration)
b	bubble
enh.	enhancement

wicking on the surfaces. In addition to the capillary wicking, the modified wettability caused by the surface coating can affect the CHF with change in re-wetting phenomenon at formation of dry patches [10].

In this regard, CHF variation due to the ATF coating has been interpreted in light of the wettability, the surface roughness, and the capillary wicking potential. As a related study, Seo et al. [11] found that the CHF of FeCrAl-coated surfaces can be enhanced up to 42% attributed to increases in surface roughness and wettability. Son et al. [12] reported that there was further potential of the CHF enhancement on Cr-coated specimens up to 79% when the coating material was deposited in a nanoscale structured shape on the microscale structured surface. They revealed that these results came from the amplified capillary wicking potential on Cr-coated specimen, where the capillary wicking potential on each specimens was quantified by observing a spreading behavior of a water droplet. This implies that for more enhancement, the ATF-functional surface needs to be structured hierarchically in nano/micro scales. Furthermore, Son et al. [13] reported that the CHF enhancement on chromia-coated specimens with forming porous structures was much more enhanced up to 88.8% than that of above studies. The porous medium secured abundant path for liquid supply in boiling phenomenon and thereby enhanced the capillary wicking, which was quantified by measuring capillary height.

In these studies, with the thermal margin notwithstanding, it is necessary to evaluate an oxidizing effect on the CHF of the ATF-coated specimens. Because physical and chemical properties of fresh surface change with operating time when the surface is exposed to normal operating environments in reactor core [14]. However, most studies with an oxidized ATF-coated cladding have focused on a material characteristic such as oxidation resistance

and a change in mechanical strength under normal [15] or accident conditions [16–18]. Only a few studies have been conducted on the CHF of the oxidized ATF-coated cladding. Lee et al. [19] showed that there was little difference of pool boiling CHF between the ZrSi₂-coated specimens before and after the pre-oxidation process. This was because the oxide formation in lumped shape on the substrate resulted in the similar contact angle and surface roughness to fresh ZrSi₂-coated specimens. In contrast, Ali et al. [20] found that the CHF of FeCrAl alloy oxidized under the PWR condition was enhanced up to 34.8% despite negligible difference in an arithmetical mean height, R_a . They revealed that the CHF enhancement was came from the increased wettability although the oxide was formed in lumped shape similar to Lee et al. [19]. Above results noted that the oxidizing effect on the CHF variation should be evaluated according to coating material and shape of the coating deposition.

In this regard, this study aims at investigating the pool boiling CHF for fresh and oxidized ATF-coated surfaces in terms of changes in surface wettability, roughness, and capillary wicking capability. DC magnetron sputtering technique is used to deposit a Cr, one of the most promising ATF cladding candidates on a substrate. The coated surface is oxidized under 400 °C dry air condition in order to simulate the aging behavior under the normal operation in the nuclear core. In the early part of this study, the basic surface characteristics, such as surface roughness, surface morphology and topology, and static contact angles, are investigated on fresh and oxidized Cr-coated surfaces. Pool boiling experiment at atmospheric pressure is conducted, and CHF data obtained are compared with an existing CHF correlation based on surface wettability and roughness. In addition, the liquid droplet spreading behavior is analyzed using a high-speed video camera in order to quantify the capillary wicking capability on such surfaces.

2. Experimental setup

2.1. Specimen preparation

Substrates were made of SS304L with a dimension $45 \text{ mm} \times 10 \text{ mm} \times 2 \text{ mm}$ in length, width and thickness, respectively. Before coating the substrate, it was ground with a sandpaper having a grit number of 800 following a single direction to exclude the difference of an initial microscale roughness. Because it has been reported that the different microscale roughness can affect the CHF value. After grinding the substrate surface, an ultrasonic cleaning in ethanol and deionized (DI) water was conducted immediately to minimize the contamination of the surfaces.

To coat Cr on the ground surface, a DC magnetron sputtering technique was introduced due to its benefits of high purity and high uniformity, and moreover, structural feature of the coated surface can be controlled by deposition parameters such as DC power, working pressure, and substrate temperature [21]. Out of these, we only controlled the substrate temperature and the others were fixed. Because it has been found that controlling the substrate temperature is easy to change surface roughness in nanoscale [22]. Target values for each deposition parameter are shown in Table 1. Among ATF candidates, Cr was used as a target material, owing to its excellent corrosion resistance, and similar thermal expansion to Zr [23]. Especially, it has been informed that coating Cr on the surface is beneficial due to the formation of a protective Cr_2O_3 layer on the top surface during corrosion and oxidation. Brachet et al. [24] revealed the stability of the Cr_2O_3 layer on the Cr-coated surface by using the physical vapor deposition technique under the 415°C steam condition in the autoclave test up to 200 days. The weight gain difference between the uncoated Zr alloy and the Cr-coated Zr alloy was continuously increasing according to the exposure day. For 200 days oxidation, the weight gain of the Cr-coated Zr alloy specimens was at least 50 times lower than that of the uncoated Zr alloy specimens. Likewise, the outer Cr_2O_3 oxide layer thickness was lower than 250 nm while 15- μm -thick zirconia outer layer was observed on the uncoated Zr alloy specimen. The Cr_2O_3 layer acted as a diffusion barrier and decreased the oxygen diffusion rate [24]. Three substrate temperatures were set for the DC magnetron sputtering process to fabricate different nanoscale roughness on the Cr-coated surfaces. According to a structure zone model, the temperature range was set below 30% of its melting point (1907°C) with a working vacuum pressure of 1.0×10^{-2} torr so as to deposit compact columnar structures with nanostructures on the ground surface [25].

In order to evaluate the effect of the oxide layer formation on CHF, the Cr-coated specimens were oxidized under a fixed condition for 20 days. Oxidation conditions set in a laboratory furnace are listed in Table 2; the oxidation temperature was set to 400°C similar to the normal operating condition; Dry air environment in ambient pressure was selected in a laboratory furnace. Although the oxidation kinetics in the selected dry environment are expected to slow down and form a thinner oxide layer than the actual reactor core environment, the oxide products and the roughness change in both environments are considered similar due to

Table 2

Oxidation conditions set in a laboratory furnace.

Oxidation parameter	Oxidation condition
Exposure environment	Dry air
Exposure temperature ($^\circ\text{C}$)	400
Exposure time (day)	20
Exposure pressure (bar)	1 (ambient pressure)

same oxidizing reaction. Under the oxidation temperature of approximately 1200°C , the stable Cr_2O_3 phase is formed on the top surface based on Cr-O phase diagram.

Collectively, we investigated six test specimens in this study: fresh Cr-coated surface at substrate temperatures of 150, 300, and 600°C (Cr-T150/300/600) and oxidized Cr-coated surface at the same substrate temperatures (OxCr-T150/300/600).

2.2. Formation of nanostructures on fresh and oxidized Cr-sputtered surfaces and their characterization

Fig. 1(a) and (b) shows the cross-sectional scanning electron microscope (SEM) images acquired by NOVA NANO SEM 450 and the energy dispersion spectrometer (EDS) mapping images of the fresh and oxidized Cr-coated specimens after a milling process through a focused ion beam (FIB). The fresh Cr-coated layer was dense and columnar-shaped without any crack and void between the Cr layer and the base material, as intended for its mechanical strength and corrosion resistance. The thickness of the fresh Cr-sputtered layer was $\sim 1.5 \mu\text{m}$ without a Cr oxide layer, whereas a thin and dark oxide layer was observed on the oxidized Cr-coated layer with a thickness of $\sim 100 \text{ nm}$.

The atomic compositions on the oxidized Cr-sputtered surfaces analyzed by EDS are shown in Fig. 2(a). The Cr-O phase equilibrium graph evaluated in previous studies indicates that Cr_2O_3 is mainly formed under the oxidation temperature of $\sim 1600^\circ\text{C}$ [16]. In addition, the oxide phase of Cr_2O_3 is expected to be produced on our surface according to the previous experimental study evaluating the high-temperature oxidation of Cr-coated Zircaloy-4 under a

Table 1
Sputtering conditions for DC magnetron sputtering.

Sputtering parameter	Sputtering condition
Target material	99.95% pure Cr
Substrate temperature ($^\circ\text{C}$)	150/300/600
Exposure time (h)	1
Working pressure (torr)	1×10^{-2}
DC power (W)	150–160
Sputtering gas	100% Ar

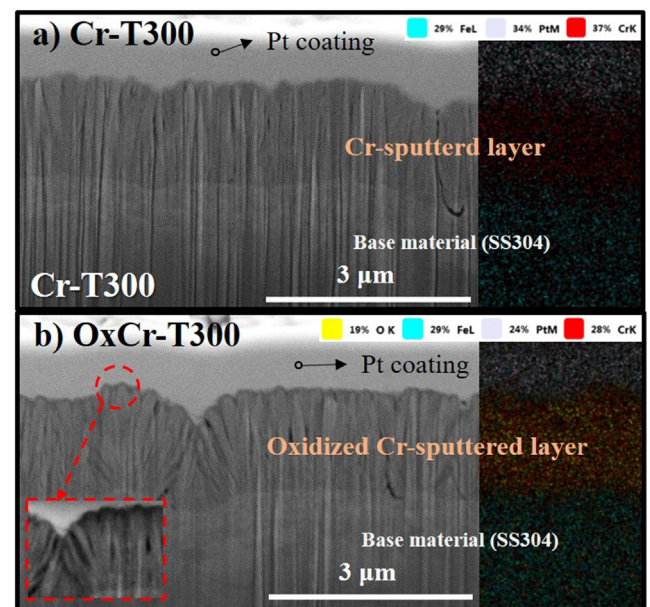


Fig. 1. Cross-sectional FE-SEM images and EDS mapping images of (a) fresh and (b) oxidized Cr-sputtered surfaces at a substrate temperature of 300°C . The dark line at the surface of oxidized surfaces in the inset image of figure (b) shows the oxide layer of the Cr layer.

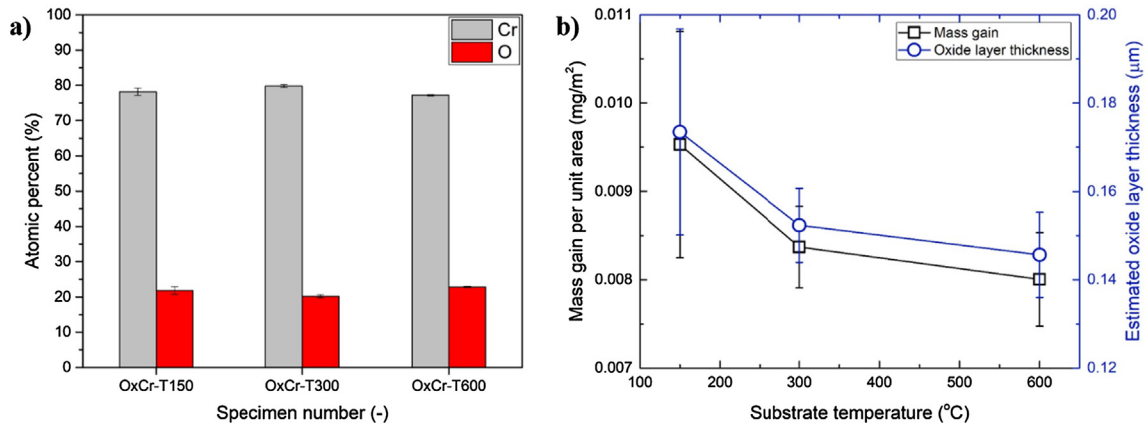


Fig. 2. Characteristics of oxidized layer: (a) composition of major elements (Cr and O) and (b) estimated oxide layer thickness.

steam environment at an oxidation temperature of 1473 K for 2000 s, where Cr_2O_3 was formed on Cr-coated surfaces [16]. An oxidation reaction of Cr is defined as



However, the results of EDS analysis of the oxidized Cr-coated surfaces confirmed that Cr was present on the surface at ~80% (O: 20%), as shown in Fig. 2(a), which is different from the element fraction of Cr_2O_3 . This difference came from the fact that the measuring beam used in the EDS detached the depth of the surface to approximately 1–2 μm and contained elemental fractions of the base coating material (Cr).

The direct measurement of the oxide layer thickness through the cross-sectional SEM images was considered to yield a large error because the thickness of the oxide layer was smaller than the resolution of the SEM images. Instead of the direct measurement, an indirect measurement method using an electronic balance of 0.01 mg resolution was adopted to evaluate the oxide thickness in this study. The oxide thickness, x , was estimated from the measured mass gain per unit area, $\Delta m/A_{ox}$, before and after the oxidation process by the following conversion equation.

$$x = \left(\frac{\Delta m}{A_{ox}} \right) \left(\frac{1}{M_{O_2}} \right) \left(\frac{2}{3} \frac{M_{ox}}{\rho_{ox}} \right) \quad (2)$$

where M_{O_2} , M_{ox} , and ρ_{ox} is the molar mass for the oxygen, the molar mass for the chromium oxide, and the density of the chromium oxide, respectively. The mole number of the inserted oxygen per unit area can be calculated as $(\Delta m/A_{ox})(1/M_{O_2})$. This is converted to the mole number per unit area of the formed chromium oxide by multiplying 2/3 because the three molecules of O_2 is consumed in order to generate the two molecules of Cr_2O_3 in Eq. (1). If the oxide layer is perfectly compact without any porous region like the fabricated layer in this study as shown in Fig. 1(b), the oxide thickness can be estimated as Eq. (2). Fig. 2(b) shows the mass gain per unit area and the estimated oxide layer thickness following the substrate temperature in the sputtering process. The scale of the estimated oxide layer thickness is similar to the directly measured oxide thickness which is about 100 nm. The thickness of the oxide layer tends to decrease as the substrate temperature increases. However, the thickness effect of the oxide film on CHF is considered negligible because the degree of oxidation (=thickness of oxide layer/thickness of coating layer) was only ~7.5% at a maximum. Thus, it was assumed that the oxidation process only affected the modification of the nanostructures on the specimen surface.

The surface topographies and morphologies were analyzed using atomic force microscopy (AFM) and SEM on microscale and

nanoscale on fresh and oxidized Cr-sputtered surfaces, as shown in Fig. 3. It is confirmed that the particulate structures on the nanoscale were formed on top of all fabricated surfaces in SEM images, same as other studies reported using the DC magnetron sputtering technique [12,13,26]. There was a significant disparity in the nanostructure shape between the fresh and oxidized surfaces. The structures on the fresh Cr-sputtered surfaces were octahedral in shape with a sharp tip, whereas the spherical particulate structures were formed on the oxidized surfaces with octahedron tips removed. The particulate structures before and after the oxidation process showed that the particle size decreased with increasing substrate temperature and the particle size on the overall Cr-coated surfaces increased after the oxidation process on all deposited surfaces. In addition, microscale scratches were formed by the unidirectional grinding process. The crucial role of the microscale scratches will be described in a later section.

All of fabricated specimen was confirmed to be more wettable than the bare specimens, based on the static contact angles measured by the Krüss EasyDrop instrument with 4 μl of DI water; the static contact angles of the fabricated surfaces were close to 0° whose value is much lower than 80.8° of the bare surface, as represented in inset images of Fig. 3. This implies that the increased contact line length due to the formation of the Cr nanostructures enhances the capillary wicking potential and liquid-solid adhesion force, causing a completely spreading behavior of the water droplet.

In analyzing the surface roughness of a structured surface, the measurement in a microscale resolution alone cannot represent the nanostructures as shown in roughness profile in Fig. 3. Therefore, a nanoscale roughness was independently quantified in order to evaluate the deposited nanostructures, which is commonly used [6,27]. On the microscale resolution of $30 \times 30 \mu\text{m}^2$, the microscratches on each fabricated surface before and after the oxidation process were quantified in terms of the arithmetical mean height, R_a , and the mean width of the profile element, R_{sm} . R_a and R_{sm} of a roughness profile were averaged in the measured area. For evaluating the nanostructures, AFM analysis in $5 \times 5 \mu\text{m}^2$ resolution was conducted to quantify the surface roughness of the nanostructures with the nanoscale surface area ratio, r_n (defined as the actual surface area to the projected surface area ratio in the measured area). The average value and the uncertainty (calculated by the standard deviation) of the roughness parameters quantified at more than 3 points on each fabricated surfaces were listed in Table 3. Additionally, changes in R_a and r_n were compared with the substrate temperature, as shown in Fig. 4. Both parameters tends to slightly decrease as the substrate temperature increases. On the other hand, the oxidation slightly augments only r_n ,

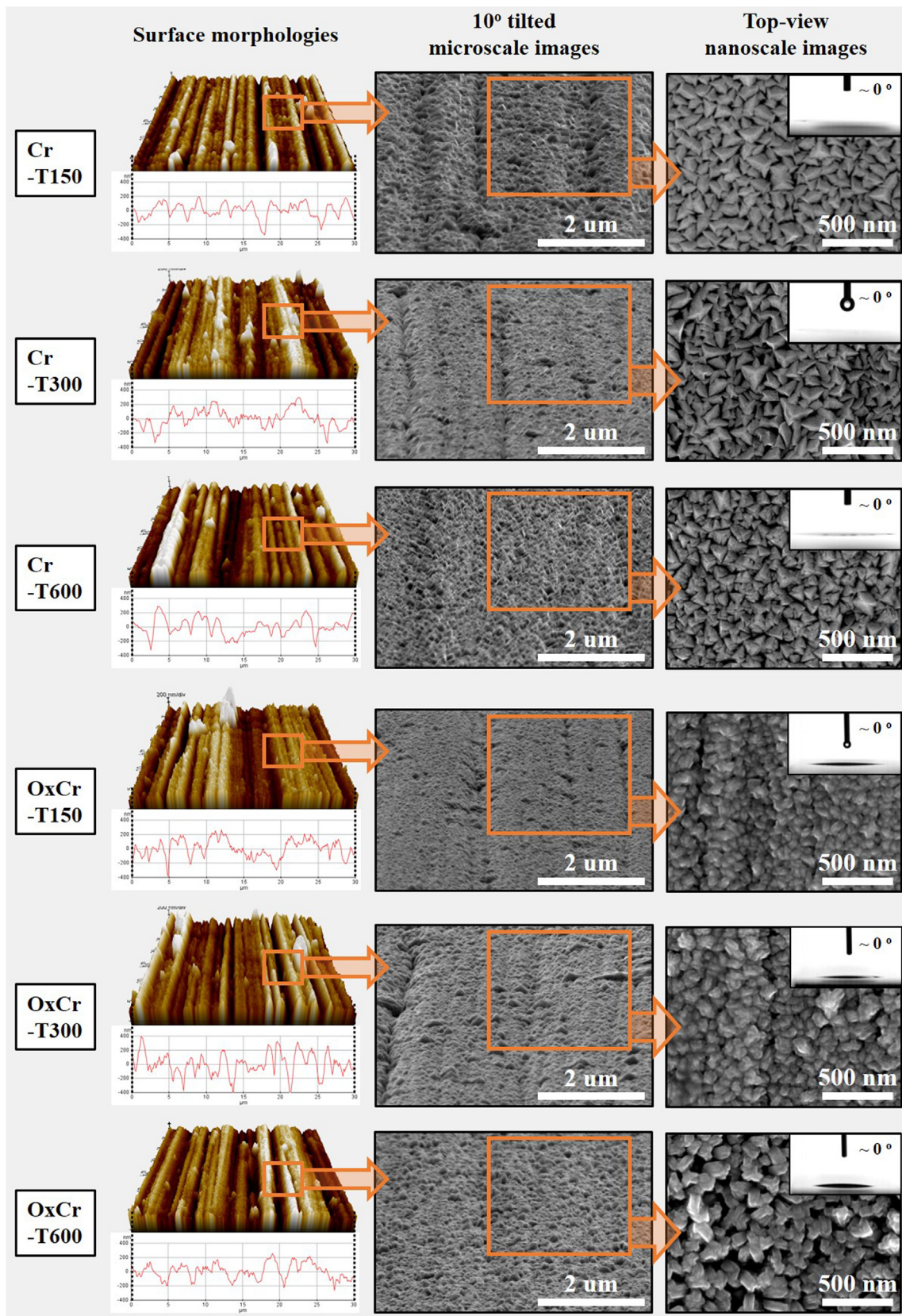


Fig. 3. Microscale surface topographies and roughness profiles measured by AFM analysis on (a) fresh and (b) oxidized Cr-sputtered surfaces. Inset images show the surface morphologies of each surface obtained by SEM analysis. The middle-sized one ($\times 20,000$) is the 10° -tilted image showing the micro-scratches and the smallest one ($\times 50,000$) is the top-view image representing the surface structures.

Table 3
Roughness parameters.

	Substrate temperature (°C)	R_a^* (nm)	R_{sm}^* (μm)	r_n^{**} (-)
Bare surfaces	–	105.27 ± 46.19	3.514 ± 1.693	–
Fresh	150	98.00 ± 9.91	1.952 ± 0.583	1.15 ± 0.03
Cr-coated	300	96.82 ± 6.04	2.202 ± 0.169	1.13 ± 0.05
surfaces	600	93.77 ± 7.71	2.605 ± 0.15	1.10 ± 0.02
Oxidized	150	106.53 ± 20.26	2.014 ± 0.377	1.19 ± 0.04
Cr-coated	300	96.37 ± 16.12	2.197 ± 0.388	1.17 ± 0.03
surfaces	600	92.94 ± 19.48	2.523 ± 0.215	1.15 ± 0.04

* Measured in the projected area of 30 × 30 μm².
 ** Measured in the projected area of 5 × 5 μm².

whereas R_a was similar before and after the oxidation, except for the Cr-T150 surface deposited with the largest size of the nanostructures. However, the difference of r_n between test surfaces seems quite small, which is close to its uncertainty represented as each standard deviation. Therefore, it is not enough to evaluate CHF enhancement with the surface roughness alone on the micro/nanostructured specimens.

2.3. Pool boiling experiment setup

Fig. 5(a) shows a schematic of the pool boiling experimental apparatus. The boiling chamber, made of polycarbonate with a dimension of 300 × 300 × 300 mm³, is filled with DI water which is heated by four immersed heaters with a total power capacity of 5 kW. The data acquisition system records the voltage and temperature history measured at test specimens. A detailed design of the test section used in the study is shown in Fig. 5(b). The dimension of the test specimen is 45 mm 10 mm, and 2 mm in length, width, and thickness, respectively. The heat-transfer surface area is 25 mm × 10 mm = 250 mm² because the remaining 10 mm length of both sides is connected with copper electrodes and copper covers in order to secure an electrical contact. The electrical

resistance of SS304L was measured to 1.0 mΩ, which is much higher than copper resistance of 0.0136 mΩ. The side and bottom of the test specimen is insulated by an epoxy of 5 mm thick filled inside a supporting zig made of a polyether ether ketone (PEEK). Three K-type thermocouples are attached underneath the specimens to measure an inner wall temperature and Pt wires are connected to the voltage taps located at each end side of the test specimen along which the voltage drop is measured.

The direct Joule heating method was used to supply the DC power to the test section. This method facilitates stepwise supply of a steady-state heat flux with a control of current input. The DC power input was controlled by a heat flux interval of 30 kW/m² and the steady state was determined by observing a steady-state surface temperature maintaining for at least three minutes. Fig. 5 (c) shows the measured voltage drop and surface temperature with time. The occurrence of CHF was judged from the surface temperature rise suddenly exceeding 200 °C in a few seconds. The applied heat flux was calculated using the following Joule's heat flux equation:

$$q'' = \frac{Power}{A_{heated}} = \frac{VI}{WL_{heated}} \tag{3}$$

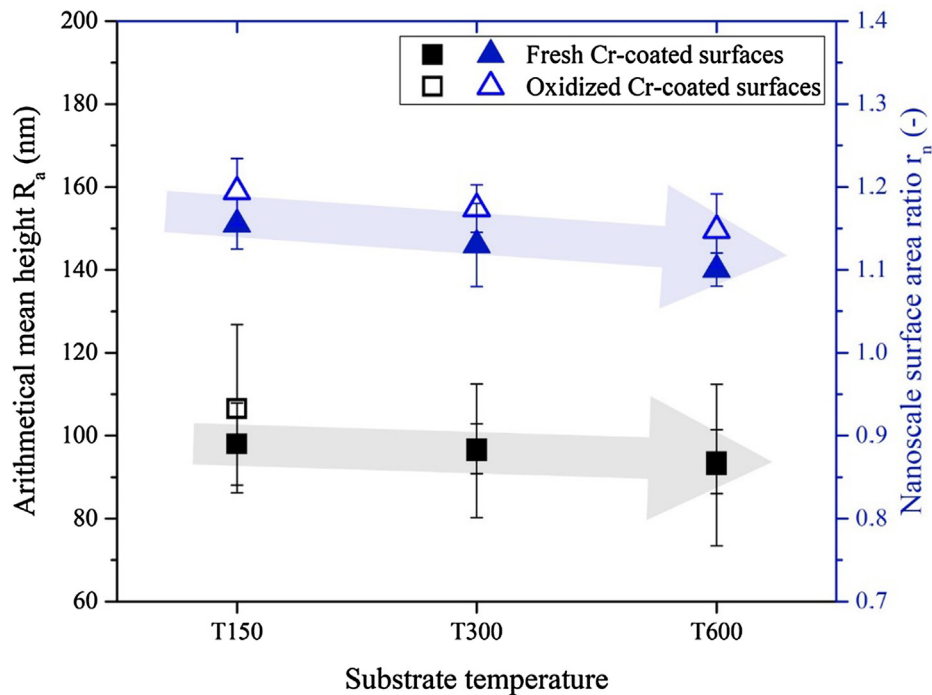


Fig. 4. Tendency of the arithmetical mean height and nanoscale surface area ratio on the fresh and oxidized Cr-coated surfaces following the substrate temperature in the coating process.

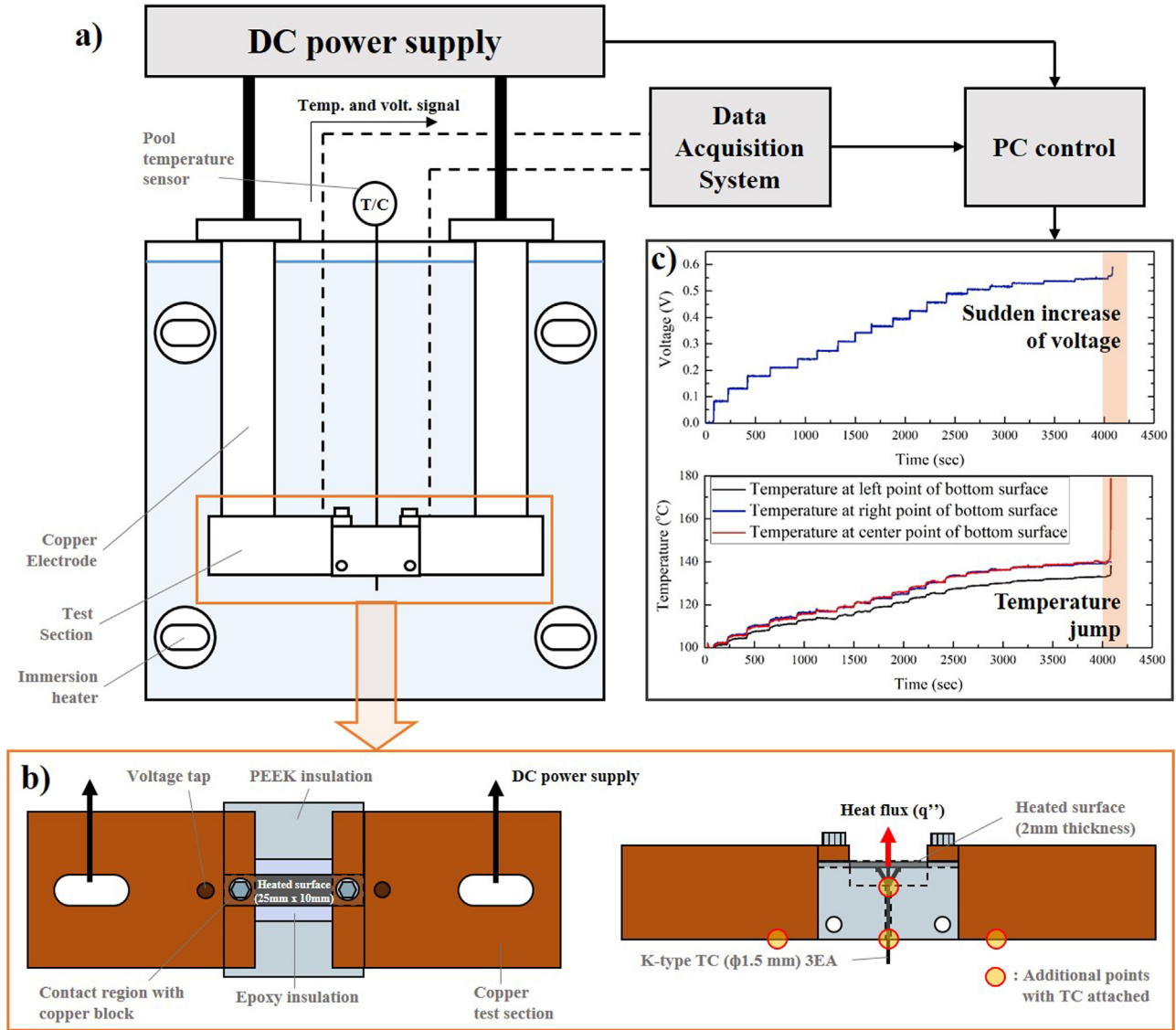


Fig. 5. Schematics of (a) pool boiling test apparatus and (b) the test section. (c) Voltage and temperature profile.

Here, V is the measured voltage drop across the test material, I is the measured current, W is the width of the heat transfer area, and L_{heated} is the heated length.

To evaluate the heat loss through the test section, four K-type thermocouples were attached on each face of the test section. The exact location of the measurement is shown in Fig. 5(b). Fig. 6 shows the temperature profiles measured at 4 different points. The maximum heat loss calculated by the 1-D thermal conduction equation was 0.56 kW/m^2 , whose value corresponds to 0.1% of the applied heat flux. The heat loss in this experiment, therefore, was neglected.

The uncertainty of the measured heat flux was estimated from the error propagation method by using the above independent variable:

$$\left(\frac{U_{q''}}{q''}\right)^2 = \left(\frac{U_V}{V}\right)^2 + \left(\frac{U_I}{I}\right)^2 + \left(\frac{U_W}{W}\right)^2 + \left(\frac{U_{L_{heated}}}{L_{heated}}\right)^2 \quad (4)$$

The uncertainties in the voltage (U_V/V), current (U_I/I), width (U_W/W), and heated length ($U_{L_{heated}}/L_{heated}$) were less than 0.5%, 0.5%, 2.3%, and 5.4%, respectively. The resulting maximum uncertainty in the measurement of the heat flux ($U_{q''}/q''$) was evaluated as 5.9%.

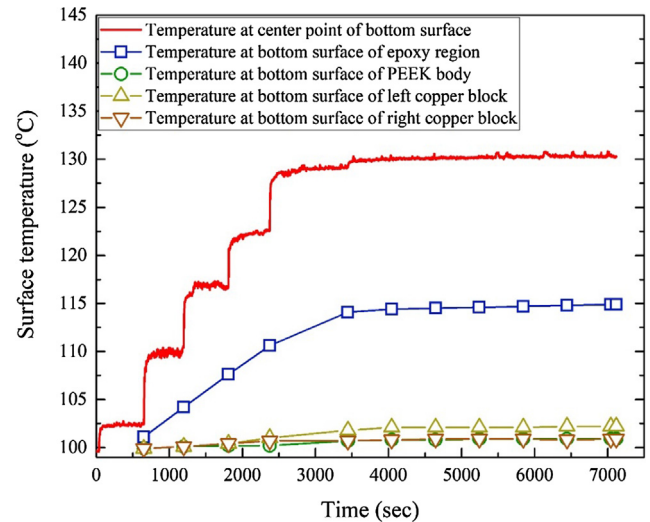


Fig. 6. Temperature profiles measured at each face of the test section.

3. Results and discussion

3.1. Coating integrity during boiling experiment.

Before evaluating the oxidation effect on the pool boiling CHF, it is necessary to ensure the coating integrity before and after the pool boiling experiments. Fig. 7 shows the photos and SEM images of the fresh and the oxidized Cr-coated surfaces, each of which was taken before boiling, after the nucleate boiling at the heat flux of 550 kW/m^2 and after the occurrence of CHF. In Fig. 7, the atomic percent of the chromium and the oxygen elements on the surfaces were evaluated by using the EDS analysis. It is confirmed that there is no meaningful change in the surface morphology and the composition before and after the nucleate boiling, while both were slightly changed after the CHF occurrence. Moreover, the oxygen element was formed more on the CHF-triggered surface than the fresh surfaces. It is because at the CHF a vapor film formed over the surface plays a role in suddenly increasing the surface temperature, which leads to rapid oxidation of the surface.

3.2. CHF enhancement trends on the fresh and oxidized Cr-coated specimens

In Fig. 8, the CHF data averaged through at least three iterative tests are compared with the substrate temperature. The maximum standard deviation of the CHF enhancement ratio was obtained as approximately 8.14% for Cr-T600 and OxCr-T300, whose value is considered reasonable in evaluating the trend of the CHF. The average CHF on the bare specimen was 705.0 kW/m^2 , which is quite smaller than the classical CHF prediction by Zuber's correlation (1108.2 kW/m^2) [28]. On the other hand, however, the measured bare CHF value is similar to the predicted value (762.8 kW/m^2) by Kandlikar's CHF correlation when a contact angle is set to 80.8° of the bare specimen [10]. Note that this study focuses on

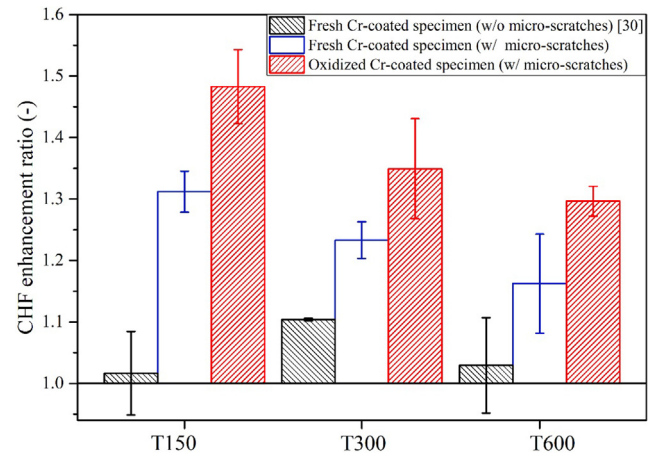


Fig. 8. Normalized pool boiling CHF results for the fresh and oxidized Cr-coated specimens.

the relative CHF enhancement after the coated specimen is oxidized.

The CHF of the fresh Cr-coated specimens, Cr-T150/T300/T600, were enhanced by 31.2%, 23.3%, and 16.3%, respectively, as compared to the bare specimens. The decreasing tendency of the CHF enhancement ratio according to the increasing substrate temperature is similar to the pool boiling CHF results on the previous Cr-sputtered specimens [12,13,29]. In addition, the CHF of the fresh Cr-coated specimen increases the substrate temperature decreases, which means that higher nanoscale specimen area ratio more enhances the CHF. Interestingly, the CHF of the oxidized Cr-coated specimens is greater than that of the fresh Cr-coated specimens; the average CHF enhancement of OxCr-T150/300/600 is 47.6%, 34.9%, and 29.7% as compared to the bare specimen. However, the enhanced CHF results on the fresh and oxidized Cr-coated

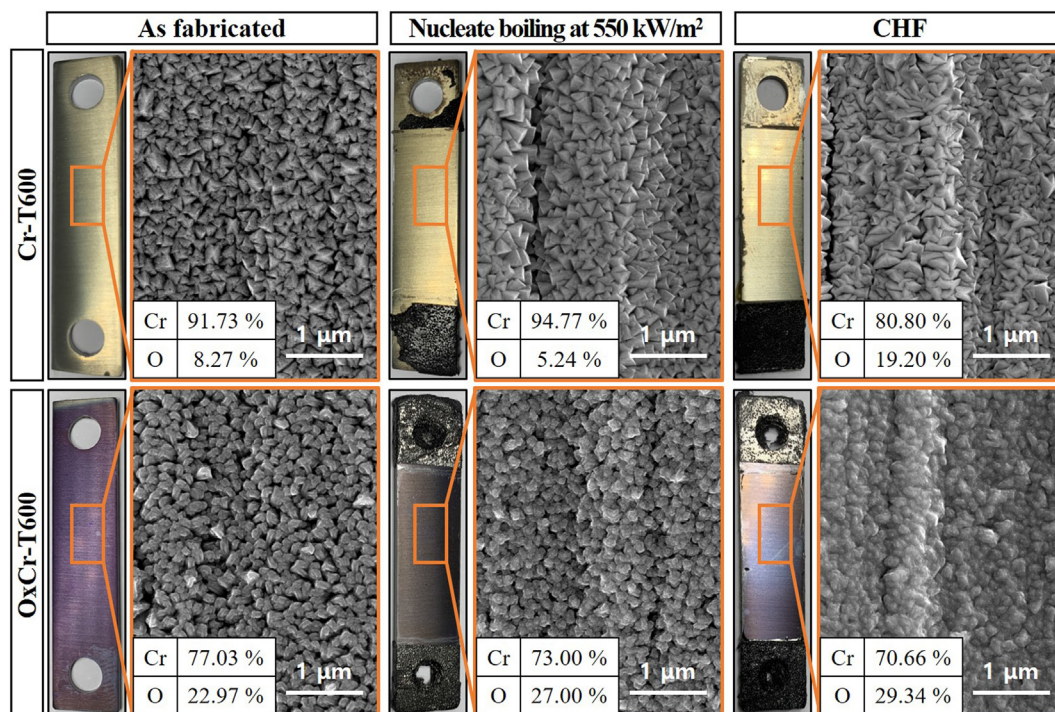


Fig. 7. Photos and SEM images of the fresh and the oxidized Cr-coated surfaces taken under conditions of as fabricated, after the nucleate boiling at 550 kW/m^2 and after the occurrence of CHF.

specimens are quite different with no CHF enhancement results on the fresh and oxidized ZrO_2 -coated specimens compared to bare specimens using same deposition technique [17], which was mentioned earlier in introduction section. This came from the lumped deposition shape of the ZrO_2 coating, which had a limit to modify the surface roughness and the capillary wicking. In addition, since ZrO_2 is a kind of oxide, there is no wettability change between the fresh and the oxidized surface.

In addition, the crucial role of the microscale scratches on the fresh Cr-coated surfaces can be discussed in comparison with the CHF obtained by Son and Kim [30] as represented in Fig. 8. Unlike current study, they deposited Cr on the mirror-polished surfaces using the DC magnetron sputtering. The CHF on the fresh Cr-coated specimen without the micro-scratches hardly increased compared to that with the micro-scratches. This comes from the characteristics of DC sputtering allowing formation of a dense layer and a smooth surface without microscale roughness. On the nanostructured smooth surface, the capillary flow was limited due to lack of the capillary channels in spite of the enhanced capillary force, which failed in delaying the formation of the irreversible dry patches. Hence, the formation of micro-scratches provides the capillary channel and the resultant wicking performance of the fabricated specimens.

As widely reported, the surface characteristics are strongly connected to the CHF. In particular, the capillary wicking phenomenon occurring on a rough surface has been known to be a dominant characteristic in enhancing the CHF [4–7,31–33]. Accordingly, recent studies have attempted to incorporate roughness parameters into the CHF enhancement mechanism. Kim et al. [7] developed a CHF model correlated with surface wettability and roughness. They simulated micro-scratches on metal surfaces to generate a capillary force, which was added as an additional force term to the existing wettability-based CHF correlation developed by Kandlikar [10].

$$q''_{\text{CHF}} = S \times h_{fg} \rho_v^{1/2} [\sigma g (\rho_l - \rho_v)]^{1/4} \times \frac{(1 + \cos \theta)}{16} \left[\frac{2}{\pi} + \frac{\pi}{4} (1 + \cos \theta) + \frac{4C \cos \theta}{1 + \cos \theta} \left(\frac{R_a}{S_m} \right) \right]^{1/2} \quad (5)$$

where q''_{CHF} , h_{fg} , σ , ρ_l , ρ_v , θ , and S are the predicted CHF, latent heat from liquid to gas, surface tension, liquid density, vapor density, static contact angle, and adjustment factor, respectively. The roughness parameters used in the capillary force term consist of the arithmetical mean height R_a and the mean spacing S_m with a proportional constant C . The pool boiling CHF experiments on the metal surfaces with R_a ranging from 0.041 to 2.36 μm were conducted to verify the developed model. The proposed model showed good agreement with the experimental data for the ground surfaces with the static contact angle of 64.5°, as shown as red dots in Fig. 9. To adopt the experimental data in this study, it is necessary to evaluate the parameter that was unused in this study (S_m). Similarly, Li and Huang [34] evaluated an empirical equation of R_a for R_a/S_m by a quadratic fitting using the data of Kim et al. [7]. Following this approach, we can calculate R_a/S_m values on the fresh and oxidized Cr-coated surfaces. In addition, S and C are set to 0.811 and 87.8 and the static contact angle was used instead of the receding contact angle, as same as Kim et al. [7]. Since the receding contact angle is lower than the static contact angle, the receding contact angles on all specimen are considered as 0° due to the superhydrophilicity of the fresh and oxidized Cr-coated surfaces. As represented by the blue shadow region in Fig. 9, the above model well predicts the CHF data on micro-structured, nanoporous [35], and micro-scratched surfaces [7,35]. However, this roughness-based CHF model hardly explain the experimental trend on the fresh and oxidized Cr-coated surfaces, showing nanostructures in this study and in the open literature [35]. According to this model, the

superhydrophilic surfaces with ~ 0.005 of R_a/S_m show a CHF enhancement ratio of approximately 2.0, while the measured CHF enhancement ratios increase by up to 1.5. In this regard, this model overestimates the increase in CHF on the superhydrophilic nanostructured surfaces, emphasizing that microscale roughness alone is insufficient to predict the CHF on the coated surface.

3.3. Increased capillary wicking on oxidized surface and its effect on CHF enhancement

As mentioned in the introduction section, researchers have tried to explain the mechanism of CHF enhancement regarding the capillary wicking potential by observing the temperature field through an infrared camera and directly visualizing the behavior of the liquid-vapor interfaces underneath nucleate bubbles [8,9,36]. They revealed that a wicked liquid layer underneath the nucleate bubble was formed around the dry spot, as depicted in Fig. 10. The wicked liquid inflow is driven by the capillary wicking pressure through a capillary channel combined with nanostructures and such inflow resists the vapor recoil pressure to suppress a growth of dry spots. In addition, the wicked liquid requires more heat flux to be evaporated, which delays the formation of the irreversible dry spot and patch. According to this observation, it is necessary to quantify the capillary wicking momentum to predict the excessive CHF enhancement, generally represented as

$$q''_{\text{enh.}} = \frac{\dot{m}_w h_{fg}}{A_t} = \frac{\rho_l u_w A_c h_{fg}}{A_t} = \left(\frac{\rho_l h_{fg}}{A_t} \right) \dot{V}_w \propto \dot{V}_w (= u_w A_c) \quad (6)$$

where $\dot{m}_w$ is the mass flow rate of the wicked liquid inflow; A_t and A_c are the heat transfer area and the total cross-sectional area of the wicked channel, respectively; u_w is the wicking velocity; and $\dot{V}_w$ is the wicked liquid volume rate. As a relevant study, Yu et al. [9] visualized the behavior of liquid-vapor interfaces underneath the nucleate bubbles on micro-pillar surfaces by using a synchrotron x-ray. From the high-resolution x-ray images, they could quantify an average capillary-induced volume rate underneath the nucleate bubble and confirmed the analogous increasing trend between the wicked liquid volume rate and the CHF enhancement ratio compared to the roughness ratio.

However, the direct measurement of the wicked liquid volume rate can be applied only on a transparent substrate, such as a silicon wafer. To quantify the capillary wicking momentum on a metal-based substrate, indirect measurements, such as the analysis of a liquid droplet spreading behavior [12,37–39], the wicking behavior using a capillary tube [27], and the receding motion of the triple contact line using an air injection [35], are needed. In this study, the spreading behavior of the liquid droplet under ambient condition is analyzed because this measurement method is the simplest and widely known out of those indirect measurement methods.

The Krüss EasyDrop instrument coupled with a high-speed camera (Phantom v7.3) is used to quantify the spreading and wicking behaviors on the test specimens (Fig. 11(a)). Generally, there are two ways to measure the spreadability on the side (Fig. 11(b)) or on the top (Fig. 11(c)). However, on the non-porous structured surfaces showing a small amount of liquid absorption onto the surfaces, it is observed that a precursor line moves almost simultaneously with a contact line of a droplet, as shown in Fig. 11(d). Thus, here is little difference between the contact diameters measured laterally and above according to the spreading times. Following this approach, the contact diameter in this study is measured from the side view so as to quantify the spreadability along the capillary channel. The frame rate of the high-speed camera is set to 2000 frames per second (fps). As an experimental procedure, a water droplet is generated at the tip of a syringe needle of

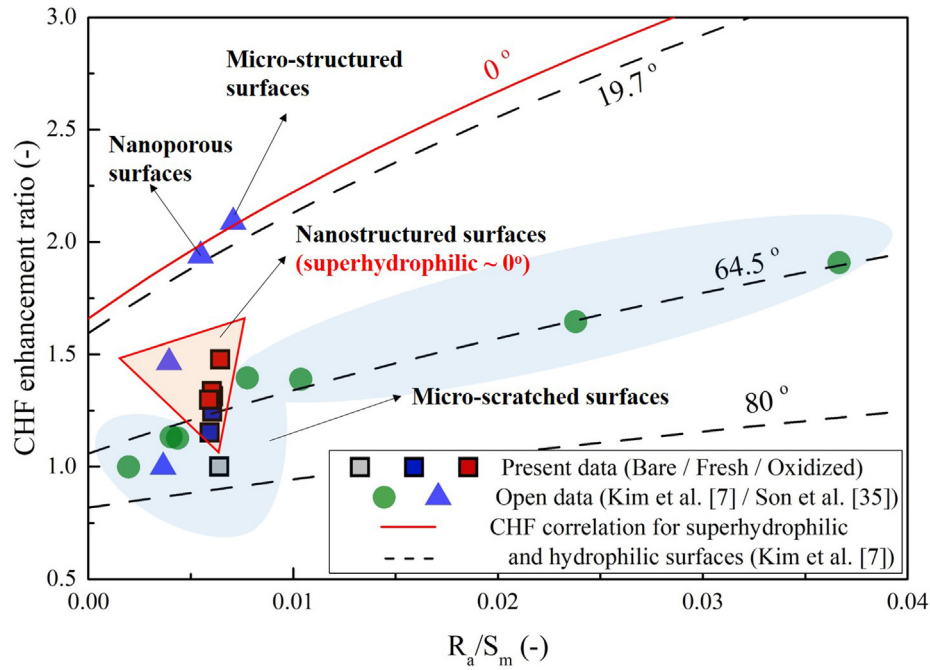


Fig. 9. Trend of CHF correlation for superhydrophilic and hydrophilic surfaces developed by Kim et al. with the measured data on the fabricated surfaces and the open data.

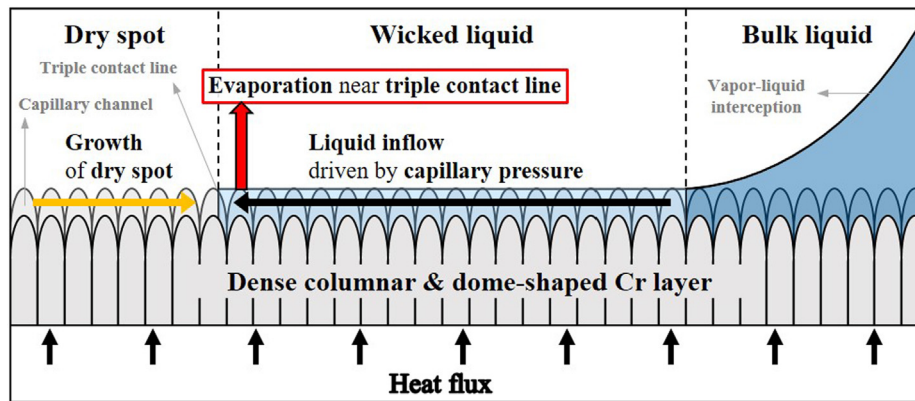


Fig. 10. Schematic of the effect of capillary wicking on CHF enhancement. Liquid inflow driven by capillary wicking pressure is drawn through capillary channel with nanostructures and conflicts with vapor recoil pressure from the growth of the dry spot.

the wetting equipment and the droplet is attached to the substrate surface moving upward slowly. The contact diameter starts to be measured over time from the moment when the water droplet comes in contact with the specimen surface. It was reported that the volume of the water droplet could affect an earlier spreading stage in the spreading behavior, where an inertia force is dominant [40]. However, the region of interest in this study is a secondary spreading stage, where the capillary wicking is much more dominant than the inertia force. The volume of the water droplet was fixed to 4 μl for all tests.

Fig. 12 shows the measured contact diameters following the spreading time on the fabricated surfaces. The droplet spreading behavior appears to be separated into two regions as reported in previous studies regarding the spreading behavior of the water droplet on the nanostructured surfaces [12,38–41]. Wemp and Carey [40] explained that the spreading behavior initially represented a synchronous spreading shape, where the wicked liquid between the nanostructures extended with the liquid droplet until the upper droplet became an intrinsic contact angle. If the

equilibrium between the apparent and the intrinsic contact angle is achieved, the expansion of the upper droplet stopped and only the wicked liquid is continuously extended, which is referred as the hemi-spreading process.

Although the spreading processes in our experiment and in the study of Wemp and Carey [40] appear similar, there is a difference in the continuous expansion of the upper-droplet contact line in the secondary region. This seems reasonable because the capillary pressure on the nanostructures of our surfaces is insufficient, and hence, the hemi-spreading process appears. Instead, the separated process observed in this experiment is induced to the dominant capillary pressure following the reduced droplet impact, similar to the spreading behavior on the other non-porous surfaces [12,38,39]. The wicking behavior of the liquid in the boiling situation is expected to be similar to the initial spreading behavior in the secondary region because there is no impact in the boiling situation. Moreover, the spreading behavior according to the different roughness is only distinguishable in the secondary region. Therefore, the spreading velocity on the fabricated surfaces is quantified

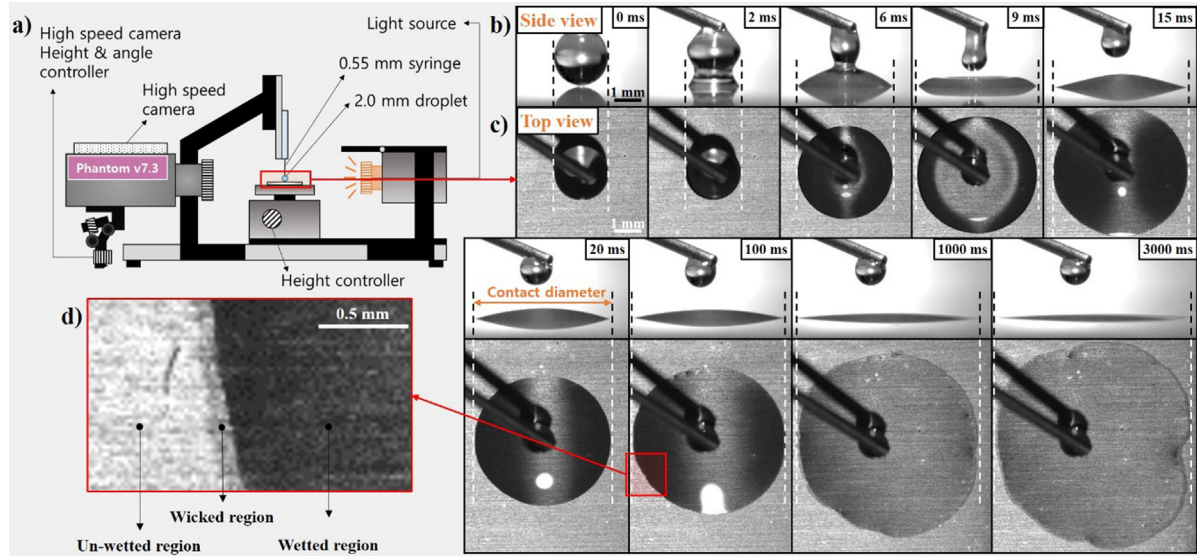


Fig. 11. (a) Schematic of Krüss EasyDrop instrument and high-speed images of liquid spreading behavior from detaching the liquid droplet to test the specimen surface: (b) side view, (c) top view, and (d) observation of wicking region between un-wetted and wetted regions.

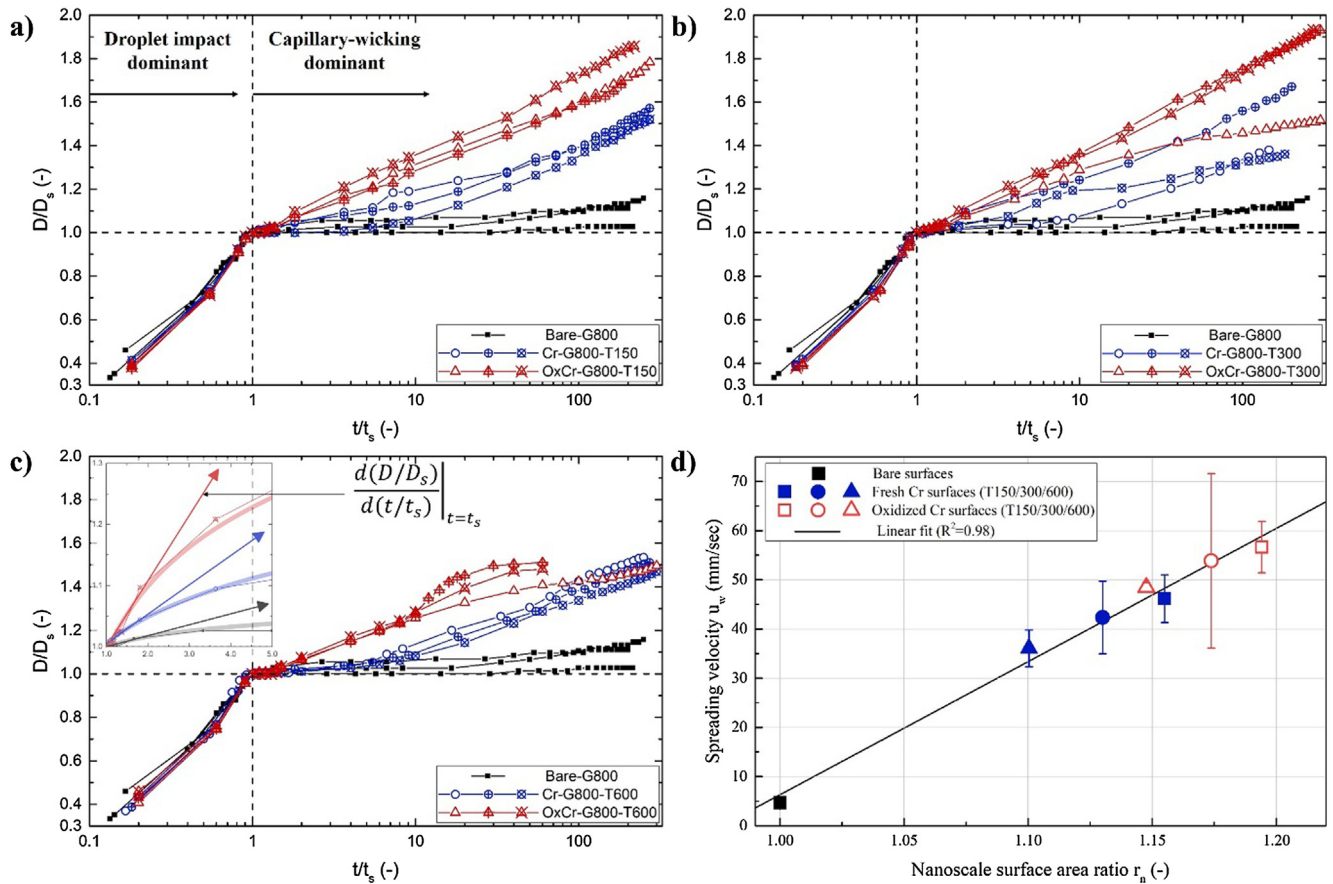


Fig. 12. Increasing behavior of a contact diameter following the spreading time normalized by the transient point of the spreading behavior on the bare, fresh Cr-coated, and oxidized Cr-coated specimens deposited at substrate temperatures of (a) 150 °C, (b) 300 °C, and (c) 600 °C. (d) Comparison of the resultant spreading velocities with the nanoscale surface area ratio on each surface.

by calculating the slope at the separation point between two regions, as shown in the inset graph of Fig. 12(c). The resultant spreading velocities on the bare, Cr-T150/300/600, and OxCr-T150/300/600 surfaces are 4.64, 46.17, 42.34, 36.08, 56.64, 53.88,

and 48.44 mm/s, respectively. We confirm that the spreading velocities can be distinguished even for the specimens with a similar contact angle. As shown in Fig. 12(d), the spreading velocity on the fresh and oxidized Cr-coated surfaces show a strong linear

relation with the increasing nanoscale surface area ratio. It can be deduced that the spreading velocity on the smaller size of the particulate nanostructures appear slower. Moreover, the spreading velocity on the surfaces coated at each substrate temperature increases after the oxidation process.

As mentioned above, the wicking momentum sucked into the dry spot underneath the nucleate bubble forms the main basis for enhancing CHF. In this regard, it is necessary to quantify the total cross-sectional area of the capillary channels, A_c , to calculate the wicked liquid volume rates ($\dot{V}_w = u_w A_c$ in Eq. (6)) on each surface from the resultant spreading velocities. Fig. 13 describes the calculating method of the cross-sectional capillary channel area. First, the cross-sectional area of the capillary channel without considering the roughness effect of the nanostructures, A_{micro} , is calculated as a simple relation $A_{micro} = 3R_a R_{sm}/2$, where R_a and R_{sm} are the aforementioned AFM data from $30 \times 30 \mu m^2$. A detailed basis of this calculation was reported by Son et al. [38]. In addition, assuming that the nanoscale surface area ratio represents the roughness of the incline of the capillary channel, the cross-sectional area in a capillary channel is calculated by multiplying r_n with A_{micro} . Moreover, the number of the capillary flow channel, N_c , is applied to calculating A_c . N_c is calculated through the circumference of the bubble base diameter, D_b , expressed as the relation of $(\pi D_b / R_{sm})$. In addition, although the unidirectional capillary channels were formed on the specimen surface, the calculation of the capillary channels through the circumference is reasonable because R_{sm} corresponding to the peak-to-peak distance of the roughness profile is much smaller than the bubble base diameter with the range between 100 nm and 200 nm. Here, it is assumed that the bubble base diameter, D_b , is identical to the Laplace length ($\sqrt{\sigma / \rho_l g}$) following the analysis of the maximum dry spot radius [6,32,35]. In this calculation process, the total cross-sectional area of the capillary flow channel, and thereby, the wicked liquid volume rate can be expressed as Eqs. (7) and (8), respectively. Finally, the substitution of Eq. (8) into Eq. (6) yields a predicted CHF correlation, as represented in Eq. (9), which comprises the roughness parameters (such as R_a , R_{sm} , and r_n) and the spreading velocity (u_w) evaluating the wicking potential of the surfaces indirectly. Here, a factor K is considered to accommodate the difference between the measured and predicted CHF enhancement values.

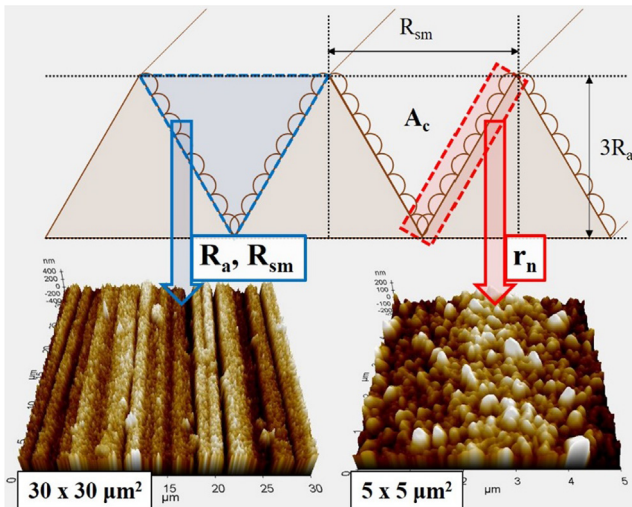


Fig. 13. Schematic of the calculation of the cross-sectional capillary channel area, which is the combination of the arithmetical mean height and mean width of profile elements from the AFM results on the scale of $30 \times 30 \mu m^2$ and the nanoscale surface area ratio from the AFM results on the scale of $5 \times 5 \mu m^2$.

$$A_c = N_c A_{micro} r_n = \left(\frac{\pi D_b}{R_{sm}} \right) \left(\frac{3}{2} R_a R_{sm} \right) r_n = \left(\frac{\pi}{R_{sm}} \right) \sqrt{\frac{\sigma}{\rho_l g}} \left(\frac{3}{2} R_a R_{sm} \right) r_n \quad (7)$$

$$\dot{V}_w = u_w A_c = u_w \left(\frac{\pi}{R_{sm}} \right) \sqrt{\frac{\sigma}{\rho_l g}} \left(\frac{3}{2} R_a R_{sm} \right) r_n \quad (8)$$

$$q''_{enh} = K \left(\frac{\rho_l h_{fg}}{A_t} \right) \dot{V}_w = K \left(\frac{\rho_l h_{fg}}{A_t} \right) u_w \left(\frac{\pi}{R_{sm}} \right) \sqrt{\frac{\sigma}{\rho_l g}} \left(\frac{3}{2} R_a R_{sm} \right) r_n \quad (9)$$

The calculated wicked liquid volume rates on the bare, Cr-T150/300/600, and OxCr-T150/300/600 following Eq. (8) are 0.1440, 1.5421, 1.3678, 1.0986, 2.1265, 1.7986, and $1.5246 \times 10^{-2} \text{ mm}^3/\text{s}$, respectively. There is a strong linear relation between the wicked liquid volume rate representing the capillary momentum and the amplified CHF on the fabricated surfaces, which is similar to the results obtained by the research correlating the spreading behavior with CHF enhancement [4,12,38,39]. Fig. 14 shows a comparison of the experimental data with the predicted CHF enhancement on the fabricated surfaces. Most of the CHF enhancement predicted from Eq. (9) is in good agreement with the experimental data, with a high R^2 value of 0.96. This result implies that the capillary wicking momentum is increased by the modified surface roughness because of the coating and the oxidation process.

Fuel cladding in the nuclear core actually has been burned for at least 1.5 years. Therefore, it is necessary to evaluate the enhancement in CHF on the oxidized Cr-coated specimens in light of the further oxidation time. As a predicted result of long-term oxidation, the surface roughness of Cr-coated specimen increases with formation of more oxide products [14], which is same trend with this study. According to aforementioned capillary wicking mechanism, CHF on long-term oxidized Cr-coated cladding would not be deteriorated in light of the enhanced surface roughness. However, because the thermal conductivity of oxide is lower than that of base metal, formation of thicker oxide layer formation may add additional thermal resistance, which may cause increase of the cladding temperature. To be clear, therefore, it is necessary to confirm the competing effect of the surface structure and the oxide layer thickness on the CHF in future.

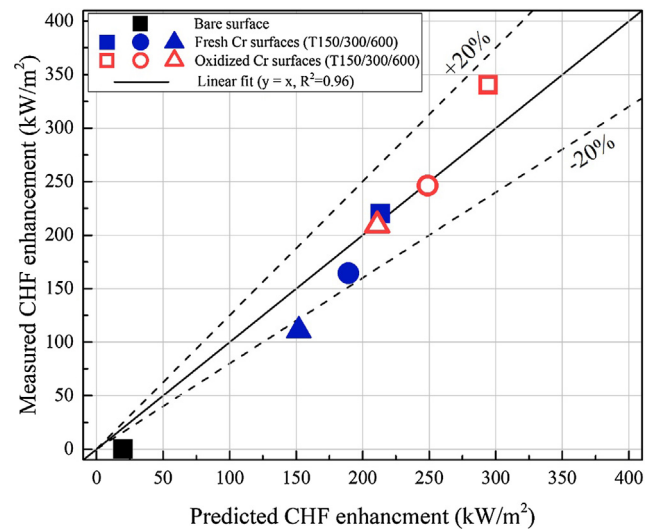


Fig. 14. Linear relation between CHF values of experimental data and the calculated wicked liquid volume rate on the bare, fresh Cr-coated, and oxidized Cr-coated surfaces.

4. Conclusion

We investigated the effect of surface structure change in the fresh and oxidized Cr-coated specimens on the liquid spreading characteristics and pool boiling CHF. A Cr layer with $\sim 1.5 \mu\text{m}$ thickness was deposited onto SS304 specimens by using a DC magnetron sputtering technique under various substrate temperatures (150, 300, and 600 °C) to control the surface roughness ($\sim 0.1 \mu\text{m}$). To simulate the normal operation in the nuclear core on the lab scale, the oxidation process was conducted at 400 °C dry air for 20 days. The resultant CHF was compared with the surface roughness (R_a , R_{sm} , and r_n), the wettability (static contact angle), and the spreadability. The major findings can be summarized as follows:

- The morphology of the fresh and the oxidized Cr-sputtered coating layer and the surface roughness were analyzed using FIB, SEM, EDS, and AFM. The particulate nanostructures were deposited on the substrate after Cr-coating process. Both the fresh and oxidized Cr-coated specimens were superhydrophilic as confirmed by static contact angle of 0°, and thereby showing the perfect spreading behavior in all cases. In addition, it was revealed that spreading velocity after the oxidation process became faster than that of the fresh coated specimens.
- Pool boiling CHF on the fresh Cr-coated specimens were enhanced by up to $\sim 31\%$, and after oxidation, the CHF was further enhanced up to $\sim 48\%$ with increased surface roughness. These CHF results showed a decreasing trend with the increasing substrate temperature during the coating process. In addition, this trend continued on the oxidized Cr-coated surfaces.
- In order to evaluate the capillary wicking potential on the fresh and oxidized Cr-coated specimens, the wicked liquid volume rate was calculated using the surface roughness parameters and the spreading velocity. The wicked liquid volume rate showed a strong linear relation with CHF enhancement, and thereby, the calculated CHF enhancement agreed well with the experimental results.
- In conclusion, this study emphasized that the oxidation behavior on the Cr-coated surfaces would not deteriorate the safety thermal margin (CHF) with enhancing the capillary wicking potential.

Declaration of Competing Interest

We declare that the submitted work has no conflict of interest with any relevant research institutes, academics, or industries.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijheatmasstransfer.2019.118655>.

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